Unit-1

Chapter-1 Introduction to ai

What is Intelligence?

Definition

Intelligence is the ability of a person or a machine to acquire knowledge, learn from experience, understand situations, think logically, solve problems, and make decisions.

Definition:
Deductive reasoning is the process of drawing a specific conclusion from general facts or rules.

Formula:
General Rule → Specific Conclusion

Example:

- **All birds have wings.**
- **A parrot is a bird.**
- **Therefore, a parrot has wings.**

Features:

- **Based on facts and rules.**
- **Gives accurate conclusions if the premises are true.**

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b) Inductive Reasoning

Definition:
Inductive reasoning is the process of making a general conclusion based on specific observations or examples.

Formula:
Specific Observations → General Conclusion

Example:

- **The sun rose in the east yesterday.**
- **The sun rose in the east today.**
- **Therefore, the sun always rises in the east.**

Features:

- **Based on observations.**
- **Conclusion may not always be 100% certain.**

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c) Analogical Reasoning

Definition:
Analogical reasoning is solving a problem by comparing it with a similar problem or situation.

Formula:
Similar Situation → Similar Solution

Example:

- **A bicycle has two wheels.**
- **A motorcycle also has two wheels.**
- **Therefore, both require balancing while riding.**

Features:

- **Uses similarities between situations.**
- **Helpful in learning and problem-solving.**

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2. Learning

Definition:
Learning is the ability to acquire knowledge or skills from experience, observation, or training and improve performance.

Example:

- **A student studies programming and writes better code after practice.**
- **An AI email filter learns to identify spam messages.**

Importance:

- **Improves knowledge.**
- **Enhances performance.**
- **Helps adapt to new situations.**

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3. Problem Solving

Definition:
Problem solving is the process of identifying a problem, analyzing it, and finding the best solution.

Steps in Problem Solving

1. **Identify the problem.**
2. **Analyze the problem.**
3. **Generate possible solutions.**
4. **Select the best solution.**
5. **Implement the solution.**
6. **Evaluate the result.**

Example:
Finding the shortest route using Google Maps.

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4. Perception

Definition:
Perception is the ability to collect, interpret, and understand information from the environment using the senses or sensors.

Example:

- **Humans recognize a friend's face.**
- **AI systems detect faces or objects in images.**
- **Self-driving cars identify roads, traffic lights, and pedestrians.**

Importance:

- **Helps understand surroundings.**
- **Supports decision-making.**
- **Improves interaction with the environment.**

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5. Linguistic Intelligence

Definition:
Linguistic intelligence is the ability to understand, use, and communicate through language effectively.

Examples:

- **Reading books.**
- **Writing essays.**
- **Speaking different languages.**

- **AI chatbots answering user questions.**

Importance:

- **Improves communication.**
- **Helps understand written and spoken language.**
- **Essential for translation and speech recognition systems.**

- # What is Artificial Intelligence (AI)?
  it refers to the ability of machines or computers to perform tasks that typically require human intelligence.these tasks involves understanding,learning,reasoning problems and even decision making
  john mc carthy is widely credited with coining the term " AI" in a proposal for the 1956 **darthmouth** summer research project on AI-workshp[he organized along with **marvin Minsky,Nathaniel rchester** and **cloude shanon**

Definition

Artificial Intelligence (AI) is a branch of computer science that develops intelligent machines and software capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem-solving, decision-making, perception, and language understanding

Or

Artificial Intelligence is the science and engineering of creating intelligent systems that can think, learn, reason, and make decisions like humans.

Different Approaches to Artificial Intelligence (AI)

Artificial Intelligence can be understood through **four main approaches**. These approaches explain how an AI system should behave or think.

1. Acting Humanly (The Turing Test Approach)

Definition

The **acting humanly** approach aims to build machines that behave like humans. The machine should perform tasks in such a way that a person cannot distinguish whether they are interacting with a human or a machine.

Requirements

- Natural Language Processing (NLP)
- Knowledge Representation
- Automated Reasoning
- Machine Learning
- Computer Vision (for visual understanding)

- Robotics (for physical interaction)

Example

- ChatGPT answering questions like a human.
- Customer service chatbots.
- Voice assistants such as Siri and Google Assistant.

Advantages

- Provides natural interaction with users.
- Useful in chatbots and virtual assistants.

Disadvantages

- Human behavior is not always logical.
- Difficult to imitate all aspects of human intelligence.

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2. Thinking Humanly (The Cognitive Modeling Approach)

Definition

The **thinking humanly** approach focuses on designing machines that think the way humans think by imitating human thought processes.

Features

- Studies human brain functions.
- Uses cognitive science and psychology.
- Learns from experience.

Example

- AI systems that learn in a way similar to students.
- Educational tutoring systems.

Advantages

- Helps understand human intelligence.
- Useful in education and cognitive research.

Disadvantages

- Human thinking is complex and difficult to model accurately.

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3. Thinking Rationally (The Laws of Thought Approach)

Definition

The **thinking rationally** approach focuses on making machines think logically and reach correct conclusions using the rules of logic.

Features

- Uses logical reasoning.
- Follows mathematical rules.
- Produces logically correct decisions.

Example

- Expert systems used in medical diagnosis.
- Mathematical theorem-proving programs.

Advantages

- Produces logical and accurate conclusions.
- Suitable for structured problem-solving.

Disadvantages

- Real-world problems often have incomplete or uncertain information.
- Pure logic cannot always handle ambiguity.

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4. Acting Rationally (The Rational Agent Approach)

Definition

The **acting rationally** approach focuses on building intelligent agents that choose the best possible action to achieve a goal based on the available information.

Features

- Makes decisions based on the current situation.
- Selects the action that maximizes success.
- Can work even with incomplete information.

Example

- Self-driving cars.
- Robot vacuum cleaners.
- Chess-playing AI.
- Navigation systems like Google Maps.

Advantages

- Flexible and practical.
- Used in most modern AI applications.
- Performs efficiently in real-world environments.

Disadvantages

- Requires good data and algorithms.
- May make incorrect decisions if information is inaccurate.

Types of Artificial Intelligence (AI)

AI is classified into **two types**:

Type-1: Based on Capabilities

This classification is based on **how capable an AI system is** compared to human intelligence.

1. Narrow AI (Weak AI)

Definition:
Narrow AI is designed to perform **one specific task**. It cannot perform tasks outside its programmed function.

Characteristics:

- Performs a single task.
- Cannot think like humans.
- Most AI systems today are Narrow AI.

Examples:

- ChatGPT
- Siri
- Google Assistant
- Face Recognition
- Spam Email Filter

Advantages:

- Fast and accurate.
- Easy to develop for specific tasks.

Disadvantages:

- Cannot perform multiple unrelated tasks.
- Has no self-awareness.

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2. General AI (AGI)

Definition:
General AI is AI that can perform **any intellectual task** that a human can perform.

Characteristics:

- Learns different tasks.
- Thinks and reasons like humans.
- Can adapt to new situations.

Examples:

- Currently, **General AI does not exist**.
- It is still under research.

Advantages:

- Can solve different types of problems.
- Learns like humans.

Disadvantages:

- Difficult to build.
- Requires advanced technology.

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3. Strong AI (Super AI)

Definition:
Strong AI, also called **Super AI**, is a hypothetical AI that is **more intelligent than humans** in every field.

Characteristics:

- Self-aware.
- Makes independent decisions.
- Has emotions and consciousness (theoretical).

Examples:

- No real-world examples.
- Exists only as a future concept.

Advantages:

- Can solve highly complex problems.
- May greatly improve science and technology.

Disadvantages:

- Ethical and safety concerns.
- Could be difficult to control.

Type-2: Based on Functionalities

This classification is based on **how AI functions**.

1. Reactive Machines

Definition:
Reactive machines respond only to the **current situation**. They do not store past experiences or learn from them.

Characteristics:

- No memory.
- Cannot learn.
- Responds only to present input.

Example:

- IBM Deep Blue Chess Computer

Advantages:

- Fast decisions.
- Simple design.

Disadvantages:

- Cannot improve with experience.

---

2. Limited Memory AI

Definition:
Limited Memory AI can **store and use past information** for a short period to make better decisions.

Characteristics:

- Learns from past data.
- Uses memory temporarily.
- Most modern AI systems belong to this type.

Examples:

- Self-driving cars
- Chatbots
- Recommendation systems

Advantages:

- Better accuracy.
- Learns from experience.

Disadvantages:

- Memory is limited.
- Depends on quality data.

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3. Theory of Mind AI

Definition:
Theory of Mind AI is an advanced AI that can understand **human emotions, beliefs, intentions, and feelings**.

Characteristics:

- Understands emotions.
- Interacts naturally with humans.
- Still under development.

Examples:

- No complete real-world examples.

Advantages:

- Better human interaction.
- Improved communication.

Disadvantages:

- Very difficult to develop.
- Ethical challenges.

---

4. Self-Awareness AI

Definition:
Self-Awareness AI is a future AI that has **its own consciousness, emotions, and self-awareness**.

Characteristics:

- Knows its own existence.
- Thinks independently.
- Makes its own decisions.

Examples:

- Does not exist today.

Advantages:

- Highly intelligent.
- Can solve very complex problems.

Disadvantages:

- Safety and ethical risks.
- Only theoretical at present.

-
- Sentiment analysis

Example

ChatGPT, Google Translate, Siri, Alexa.

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4. Computer Vision

Definition

Computer Vision enables computers to **see, analyze, and understand images and videos**.

Applications

- Face detection
- Object recognition
- Medical imaging
- Self-driving cars

Example

Face recognition used to unlock smartphones.

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5. Robotics

Definition

Robotics combines AI with engineering to build **intelligent robots** that can perform tasks automatically.

Applications

- Industrial automation
- Healthcare robots
- Space exploration
- Warehouse management

Example

Industrial robots used in car manufacturing.

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6. Expert Systems

Definition

An Expert System is an AI program that **uses the knowledge of human experts to solve specific problems and make decisions**.

Applications

- Medical diagnosis
- Financial advice
- Troubleshooting systems

Example

A medical expert system that suggests diseases based on symptoms.

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7. Artificial Neural Networks (ANN)

Definition

Artificial Neural Networks are computing systems inspired by the **human brain**. They help computers recognize patterns and make predictions.

Applications

- Image recognition
- Speech recognition
- Stock market prediction

Example

Handwriting recognition.

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8. Fuzzy Logic

Definition

Fuzzy Logic is a technique that allows AI to **make decisions using approximate or uncertain information** instead of only true or false values.

Applications

- Washing machines
- Air conditioners
- Traffic control systems

Example

An air conditioner automatically adjusts the temperature based on room conditions.

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9. Knowledge Representation

Definition

Knowledge Representation is the method of **storing information in a form that an AI system can understand and use for reasoning**.

Applications

- Expert systems
- Intelligent search engines
- Medical diagnosis systems

Example

A medical knowledge base storing information about diseases and treatments.

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10. Speech Recognition

Definition

Speech Recognition enables computers to **recognize and convert spoken language into text or commands**.

Applications

- Voice assistants
- Voice typing
- Customer support systems

Example

Google Voice Typing and Alexa.

Advantages of Artificial Intelligence (AI)

1. **Reduces human effort** – AI can perform repetitive tasks automatically.
2. **Saves time** – AI can process large amounts of data quickly.
3. **Improves accuracy** – AI can reduce errors in many tasks.
4. **Works continuously** – AI systems can work 24/7 without getting tired.
5. **Fast decision-making** – AI can analyze information and provide results quickly.
6. **Handles dangerous tasks** – AI-powered machines can perform risky jobs instead of humans.
7. **Improves productivity** – AI helps organizations complete tasks more efficiently.
8. **Personalized services** – AI can provide recommendations based on user needs.
9. **Helps in healthcare** – AI can assist doctors in diagnosis and medical research.
10. **Supports education** – AI can provide personalized learning and educational assistance.

Disadvantages of Artificial Intelligence

1. **High cost** – Developing and maintaining AI systems can be expensive.
2. **Job displacement** – Automation may reduce the need for humans in some jobs.
3. **Lack of human emotions** – AI cannot truly understand human feelings like people do.
4. **Dependence on technology** – Excessive dependence on AI can reduce human involvement and skills.
5. **Privacy concerns** – AI systems may process large amounts of personal data.
6. **Security risks** – AI can be misused for harmful or fraudulent activities.
7. **Limited creativity** – AI generates results based on learned patterns and available data.
8. **Requires large amounts of data** – Many AI systems need sufficient quality data to work effectively.
9. **Maintenance required** – AI systems need regular updates, monitoring, and maintenance.
10. **Can make biased decisions** – If training data contains bias, AI may produce unfair results

Chapter-2 intelligent agents

An **Intelligent Agent (IA)** is an autonomous system that **perceives its environment through sensors, processes the information, and acts upon the environment through actuators to achieve specific goals**.

Agents and Environment in Artificial Intelligence (AI)

Definition of Agent

An **Agent** is anything that can **perceive its environment through sensors and act upon that environment through actuators** to achieve a goal.

Example: A robot uses cameras (sensors) to detect objects and robot arms (actuators) to move them.

Definition of Environment

The **Environment** is everything surrounding the agent with which it interacts.

It provides information to the agent, and the agent performs actions that may change the environment.

Examples:

- Roads for a self-driving car.
- Chessboard for a chess-playing AI.
- A room for a robot vacuum cleaner.

Agent–Environment Interaction

```
    Environment

       │

    (Percepts)

       │

     Sensors

       │

       ▼

  +-----------------+

  | Intelligent Agent|

  +-----------------+

       │

    (Actions)

       │

    Actuators
```

│

▼

Environment

Working

1. The **environment** provides information (percepts).
2. The **agent** receives the information through **sensors**.
3. The agent processes the information and decides the best action.
4. The action is performed using **actuators**.
5. The environment changes, and the cycle repeats.

Key Concepts and Terminology in Artificial Intelligence (AI)

The following are the basic concepts used in Artificial Intelligence.

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1. Environment

Definition

The **environment** is everything surrounding an agent with which it interacts. It provides information to the agent, and the agent performs actions that may change the environment.

Examples

- Road for a self-driving car.
- Chessboard for a chess-playing AI.
- House for a robot vacuum cleaner.

Importance

- Provides input to the agent.
- Determines the agent's actions.
- Changes according to the agent's actions.

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2. Sensors

Definition

Sensors are devices used by an intelligent agent to **perceive (collect information from) the environment**.

Examples

- Camera
- Microphone
- GPS
- Keyboard
- Touch sensor
- Temperature sensor

Function

- Collects information from the environment.
- Sends the information to the agent program.

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3. Actuators

Definition

Actuators are devices that help an agent **perform actions** in the environment.

Examples

- Robot arm
- Wheels
- Speaker
- Display screen
- Printer

Function

- Executes the decision made by the agent.
- Produces an action that affects the environment.

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4. Effectors

Definition

Effectors are the parts of an agent that **carry out the physical action** decided by the agent. They are controlled by actuators.

Note: In many textbooks, the terms **Actuators** and **Effectors** are used interchangeably. However, technically, **effectors are the physical parts that interact with the environment, while actuators drive or control those parts.**

Examples

- Robot hand

- Robot wheels
- Mechanical gripper
- Drone propellers

Example

In a robot:

- **Actuator:** Electric motor
- **Effector:** Robot arm that picks up an object

---

5. Percepts and Percept Sequence

Percept

Definition

A **percept** is the information that an agent receives from the environment through its sensors at a particular moment.

Example

A self-driving car detects:

- A red traffic signal.
- A pedestrian crossing the road.

These observations are called **percepts**.

Percept Sequence

Definition

A **percept sequence** is the complete history of all percepts received by the agent from the beginning until the current time.

Example

A robot vacuum cleaner receives:

1. Wall detected.
2. Turn left.
3. Dust detected.
4. Start cleaning.

This entire history is the **percept sequence**.

---

6. Agent Function and Agent Program

Agent Function

Definition

An **agent function** is a mathematical mapping that **maps every percept sequence to an action**.

It decides **what action the agent should perform** for every possible percept sequence.

Example

Percept: "Traffic light is Red"

Action: "Stop the vehicle"

Definition

A **Reactive Agent** responds **only to the current situation**. It does not plan ahead or remember past experiences.

Characteristics

- Responds immediately to environmental changes.
- Does not store memory.
- Simple and fast.
- Makes decisions based only on current input.

Examples

- Automatic doors
- Robot vacuum cleaner
- Thermostat
- Traffic light controller

---

2. Proactive Agents

Definition

A **Proactive Agent** not only reacts to the environment but also **plans ahead and takes actions to achieve future goals**.

Characteristics

- Goal-oriented.
- Plans future actions.

- Can learn from experience.
- Makes intelligent decisions.

Examples

- Self-driving cars
- Personal digital assistants
- AI-based recommendation systems
- Smart home assistants

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3. Single-Agent System

Definition

A **Single-Agent System** consists of **only one agent** that works independently to solve a problem.

Characteristics

- One agent performs all tasks.
- No interaction with other agents.
- Simple architecture.

Examples

- Chess-playing program
- Calculator
- Robot vacuum cleaner
- Personal assistant working independently

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4. Multi-Agent System (MAS)

Definition

A **Multi-Agent System (MAS)** consists of **two or more intelligent agents** that interact, cooperate, or compete to achieve one or more goals.

Characteristics

- Multiple agents work together.
- Agents communicate with each other.
- Can cooperate or compete.
- Solves complex problems efficiently.

Examples

- Traffic management systems
- Drone swarm systems
- Online multiplayer games
- Warehouse robots
- Smart city management .

Types of Environment in Artificial Intelligence (AI)

In Artificial Intelligence, an **environment** is the surroundings in which an intelligent agent operates. Based on where the agent works, AI environments are classified into three types:

1. Artificial (Fully Digital) Environment
2. Complex Software Environment (Softbots)
3. Mixed or Physical Real-World Environment

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1. Artificial (Fully Digital) Environment

Definition

An **Artificial (Fully Digital) Environment** is a **computer-generated environment** where both the agent and the environment exist completely in digital form.

Characteristics

- Entirely virtual or simulated.
- No direct interaction with the physical world.
- Easy to control and test.
- Fast processing.

Examples

- Chess game
- Video games
- Online simulations
- Virtual reality environments

Advantages

- Safe for testing AI.
- Low cost.
- Easy to modify and reset.

Disadvantages

- Does not represent all real-world situations.

- Limited interaction with physical objects.

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2. Complex Software Environment (Softbots)

Definition

A **Complex Software Environment** is a digital environment where **software agents (Softbots)** interact with other software, websites, databases, or online services.

What is a Softbot?

A **Softbot (Software Robot)** is an intelligent software program that performs tasks automatically without physical movement.

Characteristics

- Works in software applications.
- Accesses databases and web services.
- Automates repetitive tasks.
- Communicates with other software.

Examples

- Web search engines
- Email spam filters
- Chatbots
- Banking software
- Virtual assistants

Advantages

- Saves time.
- Automates repetitive work.
- High accuracy.

Disadvantages

- Depends on software and internet.
- May fail if software changes.

---

3. Mixed or Physical Real-World Environment

Definition

A **Mixed or Physical Real-World Environment** combines the **physical world** with **digital information**. AI agents interact with real objects using sensors and actuators.

Characteristics

- Includes physical objects.
- Uses sensors and actuators.
- Environment changes continuously.
- More complex than digital environments.

Examples

- Self-driving cars
- Delivery robots
- Smart home systems
- Drones
- Industrial robots

Advantages

- Solves real-world problems.
- Automates physical tasks.
- Improves efficiency.

Disadvantages

- Expensive to develop.
- Complex due to changing environments.
- Requires advanced sensors and hardware.
- # The Concept of Rationality in Artificial Intelligence (AI)
- ## Definition
- **Rationality** is the ability of an intelligent agent to **choose the best possible action** that maximizes the chances of achieving its goal based on the available information.
- ### Simple Definition (2 Marks)
- **Rationality** is the ability of an AI agent to select the **best action** to achieve its goal in a given situation.
- ---
- # What is a Rational Agent?
- A **Rational Agent** is an intelligent agent that always performs the action that is expected to produce the **best outcome** or **maximum performance** based on its knowledge and percepts.
- **Example:**
  A self-driving car slows down when it detects a red traffic signal to ensure safety and obey traffic rules. This is a **rational action**.

Factors That Affect the Rationality of an Agent

A **rational agent** always tries to choose the **best possible action** to achieve its goal. The rationality of an agent depends on several factors.

The main factors affecting the rationality of an agent are:

1. Performance Measure
2. Percept Sequence
3. Prior Knowledge of the Environment
4. Available Actions
5. Learning Ability
6. Uncertainty in the Environment

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1. Performance Measure

Definition

A **performance measure** is the standard used to evaluate **how successfully an agent performs its task**.

Importance

- Helps the agent determine whether it is achieving its goal.
- Guides the agent in selecting the best action.

Example

A robot vacuum cleaner is judged by how clean it keeps the room while using minimum time and energy.

---

2. Percept Sequence

Definition

A **percept sequence** is the **complete history of all percepts (observations)** received by the agent from the beginning until the current time.

Importance

- Helps the agent make better decisions.
- Provides information about past experiences.

Example

A self-driving car observes traffic lights, road signs, and nearby vehicles before deciding whether to stop or move.

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3. Prior Knowledge of the Environment

Definition

Prior knowledge is the information the agent already knows about its environment before performing any action.

Importance

- Helps the agent make faster and better decisions.
- Reduces uncertainty.

Example

A navigation app already knows the road map and traffic rules before suggesting a route.

---

4. Available Actions

Definition

Available actions are all the possible actions that an agent can perform in a given situation.

Importance

- More available actions provide more options for choosing the best solution.
- The agent selects the action that best achieves its goal.

Example

A robot can:

- Move forward
- Move backward
- Turn left
- Turn right
- Stop

---

5. Learning Ability

Definition

Learning ability is the capability of an agent to improve its performance by learning from experience or past data.

Importance

- Improves decision-making over time.
- Helps the agent adapt to new situations.

Example

A spam filter learns to identify new spam emails based on previously classified messages.

---

6. Uncertainty in the Environment

Definition

Uncertainty means that the agent does not have complete or accurate information about the environment, making decision-making more difficult.

Importance

- The agent must make the best possible decision even with incomplete information.
- Increases the complexity of decision-making.

Example

A self-driving car driving in heavy rain may not clearly detect road markings due to poor visibility.

Difference Between Rational Agent and Irrational Agent

Rational Agent	**Irrational Agent**
Chooses the best possible action.	May choose an incorrect or random action.
Acts to achieve its goal.	May fail to achieve its goal.
Uses available information effectively.	Does not use available information properly.
Makes logical and intelligent decisions.	Makes illogical or poor decisions.
Maximizes performance.	Does not maximize performance.
Adapts to changes in the environment.	Has difficulty adapting to changes.
More efficient and reliable.	Less efficient and less reliable.
Example: Self-driving car stopping at a red signal.	**Example:** A robot ignoring a red signal and continuing to move.

The Nature of Environment in Artificial Intelligence (AI)

Definition

The **nature of the environment** refers to the different characteristics or properties of the environment in which an intelligent agent operates. These characteristics help determine how an AI agent perceives, thinks, and acts to achieve its goals.

Specifying the Task Environment (PEAS Framework)

Definition

The **PEAS Framework** is a method used in Artificial Intelligence to **describe and specify the task environment of an intelligent agent**.

PEAS stands for:

- **P** – Performance Measure
- **E** – Environment
- **A** – Actuators
- **S** – Sensors

It helps in designing intelligent agents by clearly defining their goals, surroundings, actions, and inputs.

Components of the PEAS Framework

1. Performance Measure (P)

Definition

A **Performance Measure** is the criterion used to evaluate how successfully an agent performs its task.

Purpose

- Measures the success of the agent.
- Helps the agent achieve its goals efficiently.

Examples

- Robot vacuum: Maximum cleanliness with minimum time and energy.
- Self-driving car: Safe driving, fewer accidents, minimum travel time.

---

2. Environment (E)

Definition

The **Environment** is the surroundings in which the intelligent agent operates.

Purpose

- Provides information to the agent.
- Changes according to the agent's actions.

Examples

- Roads and traffic for a self-driving car.
- House for a robot vacuum cleaner.
- Chessboard for a chess-playing AI.

---

3. Actuators (A)

Definition

Actuators are devices that allow the agent to perform actions in the environment.

Purpose

- Execute the decisions made by the agent.

Examples

- Wheels
- Robot arms
- Steering wheel
- Brakes
- Speaker
- Display

---

4. Sensors (S)

Definition

Sensors are devices used by an agent to collect information from the environment.

Purpose

- Observe the environment.
- Send information to the agent.

Examples

- Camera
- Microphone
- GPS
- Touch sensor
- Temperature sensor
- Keyboard

Properties of Environment in Artificial Intelligence

The **environment** in Artificial Intelligence is the surroundings in which an intelligent agent operates. The environment provides information (percepts) to the agent through **sensors**, and the agent performs actions using **actuators**.

The **properties of the environment** describe the nature of the environment and help in designing intelligent agents capable of solving problems efficiently.

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1. Discrete and Continuous Environment

Discrete Environment

Definition

A **Discrete Environment** is one in which there are a **finite (countable) number of states, actions, and percepts**. The agent moves from one state to another in separate steps.

Characteristics

- Finite number of states.
- Actions occur one at a time.
- Easy to represent mathematically.
- Easier to design AI agents.

Advantages

- Easy to understand and implement.
- Requires less computational power.
- Predictable behavior.

Disadvantages

- Cannot represent real-world situations accurately.

Examples

- Chess
- Tic-Tac-Toe
- Sudoku

Continuous Environment

Definition

A **Continuous Environment** is one in which states, actions, or time change continuously rather than in fixed steps.

Characteristics

- Infinite number of possible states.
- Continuous movement.
- Time plays an important role.
- More difficult to model.

Advantages

- Represents real-world situations.
- More realistic.

Disadvantages

- Complex calculations.
- Requires more processing power.

Examples

- Self-driving cars
- Aircraft control
- Robot navigation
- Weather monitoring

Difference Between Discrete and Continuous

Discrete	Continuous
Finite states	Infinite states
Easy to solve	Complex to solve
Step-by-step changes	Continuous changes
Example: Chess	Example: Self-driving car

---

2. Fully Observable and Partially Observable Environment

Fully Observable Environment

Definition

The agent receives **complete and accurate information** about the current state of the environment.

Characteristics

- No hidden information.
- Better decision-making.
- Less uncertainty.

Advantages

- Accurate decisions.
- Simpler AI design.

Examples

- Chess
- Tic-Tac-Toe

Partially Observable Environment

Definition

The agent receives **incomplete or limited information** about the environment.

Characteristics

- Some information is hidden.
- Requires prediction and estimation.
- High uncertainty.

Advantages

- Suitable for real-world applications.

Disadvantages

- Difficult decision-making.

Examples

- Self-driving cars
- Medical diagnosis
- Robot navigation

Difference

Fully Observable	Partially Observable
Complete information	Partial information
Low uncertainty	High uncertainty
Easier decisions	Difficult decisions
Example: Chess	Example: Self-driving car

---

3. Static and Dynamic Environment

Static Environment

Definition

The environment **does not change** while the agent is making a decision.

Characteristics

- Stable.
- No time pressure.

- Easy planning.

Advantages

- Predictable.
- Simple algorithms.

Examples

- Crossword puzzle
- Sudoku

Dynamic Environment

Definition

The environment **changes continuously** while the agent is working.

Characteristics

- Time-dependent.
- Requires quick responses.
- Environment keeps changing.

Advantages

- Suitable for real-world systems.

Disadvantages

- Difficult to control.

Examples

- Traffic systems
- Stock market
- Robot navigation

Difference

Static	Dynamic
Environment doesn't change	Environment changes continuously
Easy decisions	Fast decisions required
Example: Crossword	Example: Traffic

---

4. Single-Agent and Multi-Agent Environment

Single-Agent Environment

Definition

Only **one intelligent agent** performs the task.

Characteristics

- No competition.
- Independent operation.

Examples

- Robot vacuum cleaner
- Crossword puzzle

Multi-Agent Environment

Definition

More than one intelligent agent works together or competes.

Characteristics

- Cooperation or competition.
- Communication between agents.
- More complex.

Examples

- Chess
- Football robots
- Smart traffic systems

Difference

Single-Agent	Multi-Agent
One agent	Multiple agents
No interaction	Cooperation/Competition
Easier	More complex

---

5. Accessible and Inaccessible Environment

Accessible Environment

Definition

The agent can obtain **all necessary information** about the environment through its sensors.

Characteristics

- Complete access.
- Less uncertainty.
- Better decisions.

Examples

- Chess
- Calculator

Inaccessible Environment

Definition

The agent **cannot access complete information** about the environment.

Characteristics

- Limited information.
- Hidden objects or events.
- Requires estimation.

Examples

- Medical diagnosis
- Driving in fog
- Underwater robots

Difference

Accessible	**Inaccessible**
Complete information	Incomplete information
Easy decision	Difficult decision
Example: Chess	Example: Medical diagnosis

---

6. Deterministic and Non-Deterministic Environment

Deterministic Environment

Definition

The next state is **completely determined** by the current state and the agent's action.

Characteristics

- Predictable.
- No randomness.

Examples

- Chess
- Mathematical calculations

Non-Deterministic Environment

Definition

The result of an action **cannot be predicted with certainty** because of randomness or external factors.

Characteristics

- Uncertain.
- Multiple possible outcomes.

Examples

- Weather forecasting
- Stock market
- Traffic conditions

Difference

Deterministic	Non-Deterministic
Predictable	Unpredictable
No randomness	Randomness present
Example: Chess	Example: Weather

---

7. Episodic and Non-Episodic (Sequential) Environment

Episodic Environment

Definition

Each task is **independent**, and the current action does not affect future actions.

Characteristics

- Independent episodes.
- Simple decision-making.

Examples

- Spam email filtering
- Image recognition

Non-Episodic (Sequential) Environment

Definition

Current actions **influence future states**, so decisions must consider future consequences.

Characteristics

- Requires planning.
- Memory is important.

Examples

- Chess
- Self-driving cars
- Robot navigation

The Structure of Intelligent Agents

Definition

The **structure of an intelligent agent** refers to the internal organization of an agent that enables it to **perceive the environment, process information, make decisions, and perform actions** to achieve its goals.

An intelligent agent consists of **two main components**:

1. Architecture
2. Agent Program

   Formula:Agent = Architecture + Agent Program

   An Intelligent Agent is created by combining:

- **Architecture (Hardware)**
- **Agent Program (Software)**

The architecture provides the platform, while the agent program provides intelligence.

Formula

Intelligent Agent = Architecture + Agent Program

Example

A robot vacuum cleaner:

- **Architecture: Wheels, motors, sensors, battery, processor.**
- **Agent Program: Software that detects dirt, avoids obstacles, and plans the cleaning path.**

Together they form an intelligent agent.

Agent Function vs Agent Program

Definition

Agent Function

An Agent Function is a theoretical (mathematical) function that maps every percept sequence to an appropriate action.

Simple Definition (2 Marks):

An Agent Function is a mapping from percept sequences to actions.

Agent Program

An Agent Program is the actual software or algorithm that implements the agent function and runs on the agent's architecture (hardware).

Simple Definition (2 Marks):

An Agent Program is the software that implements the agent function and decides what action the agent should take.

Agent Function

Definition

An Agent Function specifies what action an agent should perform for every possible percept sequence.

Formula

Agent Function = Percept Sequence → Action

Characteristics

- **It is a theoretical concept.**
- **Maps percept sequences to actions.**
- **Describes the behavior of the agent.**
- **Independent of hardware.**
- **Cannot run directly on a computer.**

Example

A self-driving car observes:

- **Red traffic signal**
- **Pedestrian crossing**

The agent function decides:

Action → Apply brakes and stop.

Agent Program

Definition

An Agent Program is the implementation of the agent function using a programming language. It runs on the agent's architecture and controls the agent's actions.

Characteristics

- **Practical implementation.**
- **Runs on hardware.**
- **Written in programming languages like Python, Java, or C++.**
- **Uses AI algorithms.**
- **Produces actions based on sensor inputs.**

Example

The software inside a self-driving car receives camera data, processes it, and sends commands to the brakes and steering system.

- **GPS navigation**
- **Chess-playing AI**
- **Robot path planning**

---

4. Utility-Based Agents

Definition

A Utility-Based Agent selects the action that provides the highest utility (maximum benefit or satisfaction) among all possible actions.

Working

- **Evaluates different actions.**
- **Calculates utility value.**
- **Chooses the action with the highest utility.**

Diagram

Percepts

|

▼

Possible Actions

|

▼

Utility Function

|

▼

Best Utility

|

▼

Action

Characteristics

- **Considers multiple alternatives.**
- **Maximizes performance.**
- **Makes optimal decisions.**

Advantages

- **Produces the best outcome.**
- **Handles uncertainty effectively.**
- **More intelligent than goal-based agents.**

Disadvantages

- **Difficult to define the utility function.**
- **High computational cost.**

Examples

- **Self-driving cars**
- **Stock market trading AI**
- **Airline scheduling systems**

---

5. Learning Agent

Definition

A Learning Agent improves its performance by learning from experience and adapting to changes in the environment.

Components of a Learning Agent

1. **Learning Element – Learns from experience.**
2. **Performance Element – Selects actions.**
3. **Critic – Evaluates performance and provides feedback.**
4. **Problem Generator – Suggests new actions for learning.**

Diagram

Environment

|

▼

Performance Element

▲ |

| ▼

Learning Element

▲

|

Critic

|

Problem Generator

Characteristics

- **Learns from experience.**
- **Improves over time.**
- **Adapts to new situations.**
- **Intelligent and flexible.**

Advantages

- **Continuously improves performance.**
- **Handles changing environments.**
- **Suitable for complex tasks.**

Disadvantages

- **Requires training data.**
- **Learning may take time.**
- **More complex to develop.**

Examples

- **ChatGPT**
- **Siri**
- **Google Assistant**
- **Recommendation systems**
- **Spam filters**

1. Architecture

Definition

Architecture is the **physical platform or hardware** on which the agent program runs. It provides the resources needed to execute the agent program.

Characteristics

- Physical structure of the agent.
- Receives input from sensors.
- Executes the agent program.
- Controls actuators.

Examples

- Computer
- Robot

- Smartphone
- Drone
- Self-driving car hardware

Functions

- Provides processing power.
- Stores data and programs.
- Connects sensors and actuators.
- Executes instructions.

---

2. Agent Program

Definition

An **Agent Program** is the software or algorithm that processes percepts received from sensors and decides which action the agent should perform.

Characteristics

- Runs on the architecture.
- Implements the agent function.
- Uses AI algorithms for decision-making.
- Produces actions based on percepts.

Functions

- Processes sensor data.
- Makes intelligent decisions.
- Chooses the best action.
- Learns from experience (in learning agents).

Example

The software controlling a self-driving car that decides when to stop, accelerate, or turn.

How the Components of Agent Programs Work

Definition

An intelligent agent must represent knowledge about its environment to make decisions. The way this knowledge is stored and processed is called **knowledge representation**.

There are **three main types of knowledge representation** used in agent programs:

1. Atomic Representation
2. Factored Representation
3. Structured Representation

These representations help the agent understand the environment and choose the best action.

---

1. Atomic Representation

Definition

Atomic Representation is the simplest form of knowledge representation. In this method, the environment is represented as **individual, independent states (atoms)** without describing their internal structure.

Each state is treated as a single unit.

Characteristics

- Simplest representation.
- States have no internal details.
- Easy to implement.
- Suitable for simple problems.

Working

- The agent receives a percept.
- It identifies the current state.
- It selects an action based only on that state.

Diagram

Current State

|

▼

Atomic Representation

|

▼

Select Action

Advantages

- Easy to understand.
- Requires less memory.
- Fast decision-making.

Disadvantages

- Cannot represent relationships between objects.

- Limited flexibility.
- Not suitable for complex environments.

Examples

- Chess board represented as a single state.
- Maze-solving problem.
- Traffic light controller.

---

2. Factored Representation

Definition

In **Factored Representation**, the environment is represented using a **set of variables (attributes)** that describe each state.

Instead of treating a state as one unit, it is divided into different properties.

Characteristics

- Uses variables and attributes.
- More detailed than atomic representation.
- Easier to analyze complex situations.

Working

- The agent observes different attributes.
- It combines these attributes.
- It selects the best action based on the values.

Diagram

Environment

|

▼

State Variables

(Temperature, Speed,

Location, Time)

|

▼

Decision Making

|

▼

Action

Example

A self-driving car represents the environment using:

- Speed = 60 km/h
- Signal = Red
- Distance = 10 m
- Obstacle = Present

Based on these variables, the car decides to stop.

Advantages

- More informative than atomic representation.
- Supports efficient reasoning.
- Easy to update information.

Disadvantages

- More memory is required.
- More complex than atomic representation.

Examples

- Self-driving cars.
- Weather prediction.
- Medical diagnosis systems.

---

3. Structured Representation

Definition

Structured Representation represents knowledge using **objects and the relationships between them**.

It is the most advanced representation and helps the agent understand how different objects interact.

Characteristics

- Represents objects and relationships.
- Supports logical reasoning.
- Suitable for highly complex environments.

Working

- Identifies objects in the environment.
- Determines relationships between objects.
- Uses these relationships to make intelligent decisions.

Diagram

Objects

(Person, Car, Road)

|

Relationships

(Person crossing Road)

|

▼

Reasoning

|

▼

Action

Example

A self-driving car understands:

- Person is crossing the road.
- Traffic signal is red.
- Vehicle is approaching.

The agent combines these relationships and decides to stop.

Advantages

- Represents real-world knowledge accurately.
- Supports advanced reasoning.
- Suitable for AI applications.

Disadvantages

- Complex to design.
- Requires more memory and processing power.

Examples

- Self-driving cars.
- Medical expert systems.
- Robotics.
- Smart home systems.

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